Dhia Mokhtari

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RESEARCH INTERESTS

Machine Learning (Supervised & Unsupervised), Computer Vision, Large Language Models, Physics-Informed Neural Networks, Surrogate Modeling, Multi-objective Optimization, Knowledge Graphs, and Computational Modeling of Complex Systems.

PROFESSIONAL EXPERIENCE

Data Scientist|Engineer - Teacher Assistant - Freelance 09/2026 – Present
Le Wagon Madrid, Spain (Remote)

- Support lead lecturers during live sessions and mentor engineering cohorts in advanced Data Science and Engineering concepts, including data modeling, Deep Learning architectures, model fine-tuning, and REST APIs for containerized cloud deployments.
- Resolve real-time technical tickets and act as an escalation point for complex bugs (dependency conflicts, build failures), while auditing project blueprints to standardize deployment best practices.

R&D Machine Learning Engineer - Internship 02/2026 – 07/2026
IFPEN - IFP Energies nouvelles Rueil-Malmaison, France

- Designed then developed a **Physics-Guided Neural Network** for the prediction of electric motor loss and efficiency maps.
- Conducted a comprehensive state-of-the-art literature and domain review on low-level EV motor mechanics and existing predictive research.
- Applied expertise in experimental data processing, Deep Learning, FiLM conditioning, and analytical models.
- Designed hybrid architectures using **FiLM** and motor-design descriptors to condition predictions.
- Conducted comparative studies between classical regression, black-box neural networks, and physics-guided architectures.
- Developed a robust data pipeline to clean dirty experimental data into physically consistent datasets, enabling **zero-shot generalization** across unseen motor geometries and different power specifications.

Junior Machine Learning Engineer 02/2023 – 09/2023
GamhScan Batna, Algérie

- Performed data engineering tasks including cleaning and augmentation of a complex wheat grain imaging dataset.
- Optimized a hybrid Computer Vision architecture using **Mask R-CNN** and Transfer Learning techniques.

EDUCATION

Master Computer & Network Systems – Systèmes Autonomiques - CNS SA 09/2024 – 09/2026
Université Paris-Saclay Évry, France

- Ranked second in the M2 Autonomous Systems track.
- Relevant Coursework: Multi-objective Optimization (18/20), Deep Learning, Reinforcement Learning, Machine Learning, Game Theory (16/20), Concurrent and Distributed Systems (17/20), Multi-Agent Systems, Formal Verification, Operations Research, Data Analysis, Software Defined Networks.

Master Ingénierie des Systèmes Informatiques Intelligent - ISII 09/2023 – 07/2024
Université d'Alger 1 Alger, Algérie

License Informatiques - Systèmes Informatiques SI 09/2023 – 07/2024
Université Batna 2 Batna, Algérie

ACADEMIC PROJECTS

Self-Supervised Audio Spectrogram Transformer 11/2025 – 01/2026

- Implemented an **SSAST** pretrained on AudioSet for GTZAN genre classification, outperforming CNN baselines on low-resource data.
- Engineered a robust data augmentation pipeline (**SpecAugment**, dynamic 10s sampling) to prevent overfitting.
- Optimized ViT-Base training under strict memory constraints using **Gradient Checkpointing**, **Gradient Accumulation**, **Layer-wise Learning Rate Decay (LLRD)**, and Label Smoothing.

Reinforcement Learning Framework 12/2025 – 01/2026

- Developed an RL system comparing **SARSA** and **Q-Learning** in Gymnasium, reaching theoretical optimality for the Blackjack environment.
- Engineered a **"True Skill" evaluation pipeline** and policy heatmaps to isolate agent maturity from exploration noise and visualize decision-making across 1,500+ states.
- Built a seed-synchronized **CLI benchmarking tool** to quantitatively compare human strategies against AI-optimized policies under identical stochastic conditions.

Multi-Agent Autonomous Coordination System 12/2025 – 01/2026

- Multi-agent framework in Java/Processing for competitive simulation, optimizing agent behaviors for resource logistics and tactical combat.
- Engineered a collaborative scouting protocol using targeted message passing and dynamic production queues to balance team economy and threat response.
- Designed hierarchical state-machines for agent autonomy, implementing complex behaviors such as stigmergic perimeter defense and predictive kinematics-based aiming.

Formal Modeling and Resilience of Ecosystems 09/2025 – 12/2025

- Reproduced qualitative ecosystem resilience simulations using the EDEN model and the `ecco` tool.
- Recommended by Prof. Franck Pommereau for experimental rigor and analysis quality.

Aligning LLMs with Knowledge Graphs (M1 Research Project) 01/2025 – 05/2025

- Investigated the integration of structured Knowledge Graphs (Neo4j) with Large Language Models to improve factual accuracy in generative tasks.
- Developed a Retrieval-Augmented Generation (RAG) pipeline to ground LLM outputs, effectively mitigating hallucinations for domain-specific educational content.
- Designed prompts and evaluation metrics to ensure the generated explanations were contextually accurate and adapted to specific pedagogical needs.
- **Technologies:** Python, LangChain, LangGraph, Neo4j, ChromaDB, Ollama, Pydantic.

Evolutionary Game Theory & Spatial Simulation 04/2025 – 05/2025

- Engineered a vectorized spatial simulation engine using **NumPy** and **SciPy** convolutions to model emergent cooperation dynamics via the Spatial Prisoner's Dilemma.
- Programmed a library of advanced game theory strategies (e.g., Contrite Tit-for-Tat).
- Developed an interactive GUI for real-time monitoring of population dynamics, strategy phase transitions, and statistical cluster analysis.

TECHNICAL SKILLS

Scientific ML & Surrogate Modelling: physics-guided neural networks, surrogate modelling, FiLM conditioning, engineering prediction, scientific machine learning, experimental data, model generalisation and uncertainty/OOD analysis concepts

Machine Learning & Deep Learning: Python, PyTorch, Scikit-learn, TensorFlow, Hugging Face Transformers, regression, classification, Deep Learning, computer vision, representation learning, model training and evaluation

Model Evaluation & Research: experimental design, quantitative model comparison, validation protocols, robustness, out-of-distribution evaluation, error analysis, generalisation assessment and reproducible experimentation

Data Processing: Python, Pandas, NumPy, SQL, Apache Spark, data preparation, feature engineering, data quality control, reproducible processing pipelines

Software Engineering: Python, Java, C, Git/GitHub, Linux/Bash, Docker, CI/CD, modular architectures, testing, technical documentation and reproducible workflows

MLOps & Computing: MLflow, Docker, Kubernetes, Git, experiment tracking, model lifecycle management, containerisation and deployment concepts

Generative AI: Transformers, LangChain, RAG, embeddings, knowledge graphs, Neo4j and AI assistants

Optimisation: constrained optimisation, operations research, linear programming and multi-objective optimisation

LANGUAGES & INTERESTS

Languages: English (C2 – IELTS Academic), French (C2), Arabic (Native), Tamazight - Chaoui (Native)

Interests: Cinema, Motorsports, Mechanics, Photography, Drawing, Video Games, Manga